



Writing large numbers in standard form

Task A: Writing large numbers in standard form

1) Without using a place value chart, convert the following into **standard form**.

- a) $15\,000 = 1.5 \times 10^4$
- b) $566\,000\,000 = 5.66 \times 10^8$
- c) $9\,800\,000\,000 = 9.8 \times 10^9$
- d) $156\,100 = 1.561 \times 10^5$
- e) $563\,000\,000\,000\,000\,000 = 5.63 \times 10^{17}$
- f) $100\,200\,000\,000 = 1.002 \times 10^{11}$

2) Three pupils were given $905\,000\,000\,000$ to convert to **standard form**. Identify where they made their error and give the correct answer.

$905\,000\,000\,000$

Answer: 905×10^9

The first part of **standard form** must have a number between 1 and 10, including 1 and not including 10. ($1 \leq A < 10$)

$905\,000\,000\,000$

Answer: 9.5×10^{11}

In order to correctly obtain $905\,000\,000\,000$, there must be a multiplication of powers of 10 to 9.05 not 9.5
Correct: 9.05×10^{11}

$905\,000\,000\,000$

Answer: 9.05×10^{10}

Incorrectly counted the number of multiplications of 10. There are 11 multiplications of 10.

3) Explain why **standard form** is important.

- We write a number in **standard form** because it is easier, concise and more efficient to write numbers in this way.
- It reduces errors. It would take too long to write huge numbers and there would be more room for error when writing a number in its ordinary form.
- **Standard form** allows any number to be written using one standard approach i.e. where $A \times 10^n$ and $1 \leq A < 10$ and n is an integer.
- When writing in a concise form, it is easier to use in calculations.



Task B: Standard form to ordinary

1) Write the following as ordinary numbers.

a) $1.8 \times 10^3 = 1\,800$

e) $8.02 \times 10^3 = 8\,020$

b) $1.39 \times 10^5 = 139\,000$

f) $5.001 \times 10^5 = 500\,100$

c) $3.93 \times 10^7 = 39\,300\,000$

g) $1.2345 \times 10^7 = 12\,345\,000$

d) $4.413 \times 10^8 = 441\,300\,000$

h) $7.00004 \times 10^8 = 700\,004\,000$

Task C: Standard form on the calculator

1) Fill in the missing information.



2) Investigate when the number becomes so big that the calculator converts it into **standard form**.

*Any integer on the screen which has more than 9 digits will be converted into **standard form**.*

